

Proving segment relationships

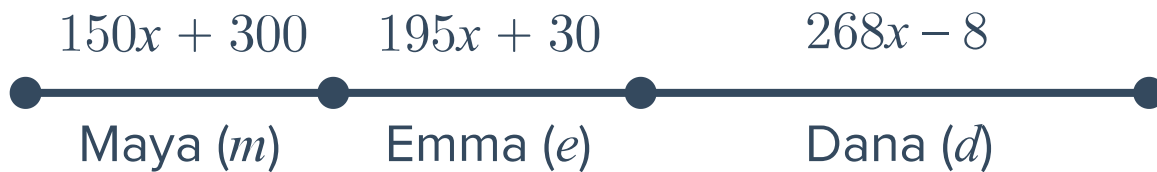


- 1 Determine whether each of the following statement is sometimes, always, or never true.
 - a The segments $\overline{AB}$ and $\overline{BA}$ describe the same set of points.
 - b Given A and B are two different points, the rays $\overrightarrow{AB}$ and $\overrightarrow{BA}$ describe the same set of points.
 - c Given A and B are two different points, line $\overleftrightarrow{AB}$ and line $\overleftrightarrow{BA}$ describe the same set of points.
 - d Given A , B , and C are three different points, the rays $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are opposite rays.
- 2 Given line $\overleftrightarrow{AB}$ passes through point C , is it true that C must also lie on $\overrightarrow{AB}$?
- 3 Suppose that points A , B , and C are collinear, with point B between points A and C .



Solve for x if $AC = 21$, $AB = 15 - x$, and $BC = 4 + 2x$.

- 4 Maya, Emma and Dana are running in a relay race. Each will run a segment of the track until the last member gets to the finish line. The total length of the race (t) is 4000 m. Maya will run $150x + 300$ m, Emma will run $195x + 30$ m, and Dana will run $268x - 8$ m.

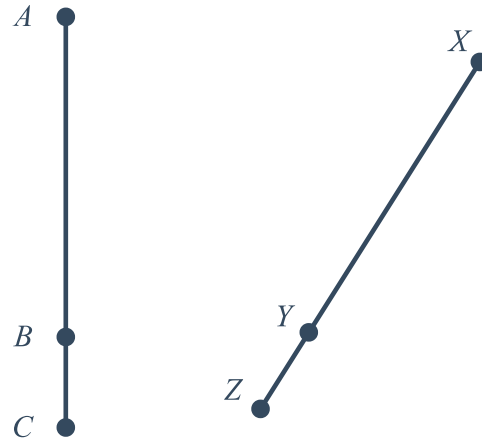


Using the information provided, solve for x .

- 5 Given: P , Q , R , and S are collinear
Prove: $PQ = PS - QS$



- 6 In the diagram shown, A , B , and C are collinear, and X , Y , and Z are collinear.
Given: $\overline{AB} \cong \overline{XY}$ and $\overline{BC} \cong \overline{YZ}$
Prove: $\overline{AC} \cong \overline{XZ}$



- 7 Given: B is the midpoint of $\overline{AC}$
Prove: $2AB = AC$



- 8 In the diagram shown, points R , S , T , and U are collinear.
Given: $\overline{RT} \cong \overline{SU}$
Prove: $\overline{RS} \cong \overline{TU}$

